Machine Learning

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The VC (Vapnik-Chervonenkis) dimension measures the capacity or complexity of a model or hypothesis class in machine learning. It quantifies how well a model can fit a variety of datasets. Specifically, the VC dimension represents the maximum number of data points a model can shatter, meaning it can perfectly classify all possible labelings of those points. A higher VC dimension indicates that the model is more flexible and capable of learning complex patterns. However, it also suggests a higher risk of overfitting, as the model might capture noise instead of meaningful relationships. Conversely, a low VC dimension suggests limited capacity, which might lead to underfitting. The VC dimension helps balance the trade-off between model flexibility and generalization. It's a theoretical tool often used in statistical learning theory to evaluate a model's generalization ability to unseen data.

Agenda:
1 Generalization Error.
2 Effective Number of Hypotheses.
3 The VC Dimension.
4 Approximation-Generalization Tradeoff.

Generalization Error

We have already discussed the value of E_{in} does not always generalize to a similar value of E_{out}. Generalization is a key issue in learning. One can define the *generalization error* as the discrepancy between E_{in} an E_{out}. The Hoeffding Inequality provides a way to characterize the generalization error with a probabilistic bound,

$$P(|E_{\mathsf{in}}(g) - E_{\mathsf{out}}(g)| \geq \epsilon) \leq 2Me^{-2\epsilon^2 N},$$

for any $\epsilon > 0$, where $g \in \mathcal{H}$, $\mathcal{H} = \{h_1, h_2, \ldots, h_M\}$. Pick a tolerance level δ, for example $\delta = 0.05$, and assert with probability at least $1 - \delta$

$$E_{\mathsf{out}}(g) \leq E_{\mathsf{in}}(g) + \sqrt{\frac{1}{2N} \log \frac{2M}{\delta}}.$$

We call this equation *generalization bound* because it bounds E_{out} in terms of E_{in}.

Effective Number of Hypotheses

The *growth function* quantifies the *effective* number of hypotheses. The growth function replaces in the generalization bound. It can handle hypothesis spaces of infinite cardinality.

Let $X_1, X_2, \ldots, X_N$ be samples from a data generation function. The *dichotomies generated* by a hypothesis space $\mathcal{H}$ on these samples are defined by

$$\mathcal{H}(X_1, X_2, \ldots, X_N) = \{(h(X_1), h(X_2), \ldots, h(X_N)) \,|h \in \mathcal{H}\}.$$

One can think of the dichotomies $\mathcal{H}(X_1, X_2, \ldots, X_N)$ as a set of hypotheses just like $\mathcal{H}$, except that the hypotheses are seen through the eyes of N points only. A larger $\mathcal{H}(X_1, X_2, \ldots, X_N)$ means that $\mathcal{H}$ is more diverse, getting more dichotomies on $X_1, X_2, \ldots, X_N$. The growth function is based on the number of dichotomies.

The growth function is defined for a hypothesis set set $\mathcal{H}$ by

$$m_{\mathcal{H}}(N) = \max_{X_1, X_2, \ldots, X_N} |\mathcal{H}(X_1, X_2, \ldots, X_N)|$$

where $|\cdot|$ denotes a set's cardinality (number of elements).

In words, $m_{\mathcal{H}}(N)$ is the maximum number of dichotomies generated on $\mathcal{H}$ any N points. To compute $m_{\mathcal{H}}(N)$, we consider all possible choices of N points $X_1, X_2, \ldots, X_N$ and pick the ones that give us the most dichotomies. Like M, $m_{\mathcal{H}}(N)$ measures the number of hypotheses in $\mathcal{H}$, except that a hypothesis is now considered on N points. For any $\mathcal{H}(X_1, X_2, \ldots, X_N) \subseteq \{-1, +1\}^N$ (the set of all possible dichotomies on any N points), the value of $m_{\mathcal{H}}(N)$ is at most $|\{-1, +1\}^N|$, hence

$$m_{\mathcal{H}}(N) \leq 2^N.$$

If $\mathcal{H}$ is capable dichotomies on of generating all possible X_1, X_2, ..., X_N, then $H(X_1, X_2, \ldots, X_N) = \{-1, +1\}^N$, and we say the $\mathcal{H}$ shatters X_1, X_2, ..., X_N. This signifies that $\mathcal{H}$ is as diverse as this sample can be.

If no data set of size $\mathcal{H}$ can shatter k, then it is said to be a *breakpoint* for $\mathcal{H}$. If the condition of $m_{\mathcal{H}}(N) = 2^N$ breaks at any point, we can bound $m_{\mathcal{H}}(N) = 2^N$ for all values of N by a simple polynomial based on this breakpoint.

Definition 1: The VC Dimension

The Vapnik-Chervonenkis dimension of a hypothesis set $\mathcal{H}$, denoted by $d_{VC}(\mathcal{H})$, is the largest N for which $m_{\mathcal{H}}(N) = 2^N$. If $m_{\mathcal{H}}(N) = N$ for all N, then $d_{VC}(\mathcal{H}) = \infty$.

If $d_{VC} = \infty$ is the VC dimension of $\mathcal{H}$, then $k = d_{VC} + 1$ is breakpoint for $m_{\mathcal{H}}(N) = 2^N$ since $m_{\mathcal{H}}(N)$ cannot be equal to 2^N for any $N > d_{VC}$ by definition. It is easy to see that no smaller breakpoint exists sin ce $\mathcal{H}$ can shatter d_{VC} points; hence, it can shatter any subset of points.

Note that if $m_{\mathcal{H}}(N) \leq 2^k$ for some value k, then

$$m_{\mathcal{H}}(N) \leq \sum_{i=0}^{k} \binom{N}{i}$$

for all N. The RHS is polynomial in N of degree $k-1$. Since $k = d_{VC}+1$ is a breakpoint for $m_{\mathcal{H}}(N)$,

$$m_{\mathcal{H}}(N) \leq \sum_{i=0}^{d_{VC}} \binom{N}{k}$$

From these results, we can show that

$$m_{\mathcal{H}}(N) \leq N^{d_{VC}} + 1.$$

The VC Generalization Bound

For any tolerance $\delta > 0$,

$$E_{\mathsf{out}} \leq E_{\mathsf{in}} + \sqrt{\frac{8}{N} \log \frac{4m_{\mathcal{H}}(2N)}{\delta}}$$

with probability $\geq 1-\delta$

Another valid bound can be obtained using the d_{VC} polynomial bound.

$$E_{\mathsf{out}} \leq E_{\mathsf{in}} + \sqrt{\frac{8}{N} \log \frac{4((2N)^{d_{VC}}+1)}{\delta}}$$

We can write

$$E_{\mathsf{out}} \leq E_{\mathsf{in}} + \Omega(N, \mathcal{H}, S)$$

where

$$\Omega(N, \mathcal{H}, S) = \sqrt{\frac{8}{N} \log \frac{4m_{\mathcal{H}}(2N)}{\delta}},$$

or

$$\Omega(N, \mathcal{H}, S) = \sqrt{\frac{8}{N} \log \frac{4((2N)^{d_{VC}}+1)}{\delta}}.$$

Approximation-Generalization Tradeoff

There is another way to look at the approximation-generalization tradeoff that is particularly suited for squared error measures rather than the binary error used in the VC analysis.

The bias-variance decomposition of the out-of-sample error is based on squared error measures.

As an example, consider the case of a polynomial regression. Let $\mathcal{D} = (\{x_1, t_1\}, \{x_2, t_2\}, \ldots \{x_N, t_N\})$ an i.i.d. sample (training set) where

$$x \sim p(x),$$

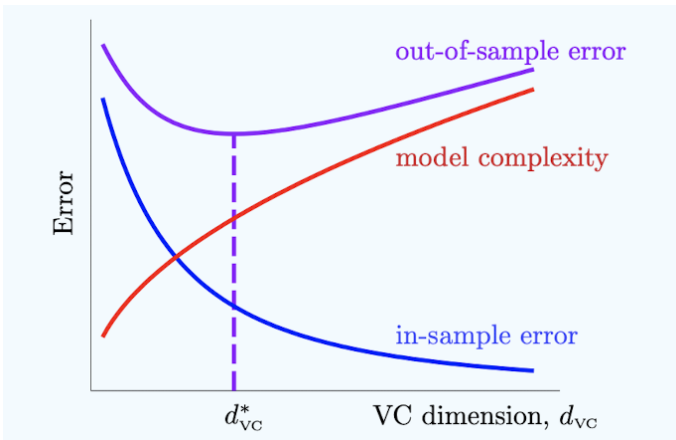


Figure 1: Learning curves. From Learning from Data.

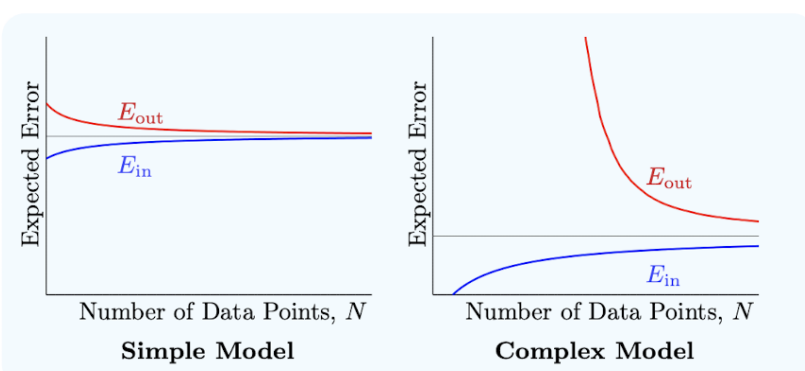


Figure 2: Learning curves. From Learning from Data.

and

$$t = f(x) = \sin(2\pi x) + \epsilon(x),$$

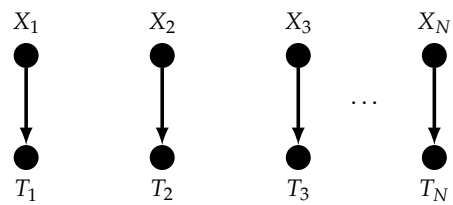
with

$$\epsilon(x) \sim \mathcal{N}(x \mid 0, \sigma^2),$$

Therefore:

$$t \mid x \sim \mathcal{N}(x \mid \sin(2\pi x), \sigma^2).$$

The graphical model associated with the sample is:



The conditional likelihood of the model is

$$p(t_1, t_2, \ldots, t_N \mid x_1, x_2, \ldots, x_N) = \prod_{n=1}^{N} \mathcal{N}(\sin(2\pi x_n) \mid 0, \sigma^2),$$

and the joint distribution of the sample is:

$$p(\mathcal{D}) = p(t_1, t_2, \ldots, t_N, x_1, x_2, \ldots, x_N) = \prod_{n=1}^{N} \mathcal{N}(\sin(2\pi x_n) \mid 0, \sigma^2) p(x_n),$$

and the joint distribution over M training sets is

$$p(\mathcal{D}_M) = \prod_{m=1}^{M} p(t_{m1}, t_{m2}, \ldots, t_{mN}, x_{m1}, x_{m2}, \ldots, x_{mN}) = \prod_{m=1}^{M} \prod_{n=1}^{N} \mathcal{N}(\sin(2\pi x_{mn}) \mid 0, \sigma^2) p(x_{mn}).$$

Assume we conducted an MLE estimation of the polynomial coefficients based on the data sample to get a specific solution $g(x)$ from the hypotheses space $g \in \mathcal{H}$. Clearly,

$$t = g(x) = g(x; \underbrace{t_1, t_2, \ldots, t_N, x_1, x_2, \ldots, x_N}_{\sim p(\mathcal{D})}).$$

By definition, the out-of-sample error is:

$$\begin{aligned} \mathrm{E}_{\mathsf{out}}(g^{\mathcal{D}}) &= \mathrm{E}_x[(g^{\mathcal{D}}(x) - f(x))^2] \\ &= \mathrm{E}_x[g^{\mathcal{D}}(x)^2] - 2\mathrm{E}_x[g^{\mathcal{D}}(x) f(x)] + E_x[f(x)] \end{aligned}$$

and the expected value of $\mathrm{E}_{\mathsf{out}}(g^{\mathcal{D}})$ with respect to $p(\mathcal{D})$ is

$$\begin{aligned} \mathrm{E}_{\mathcal{D}}\left[\mathrm{E}_{\mathsf{out}}(g^{\mathcal{D}})\right] &= \mathrm{E}_{\mathcal{D}}\left[\mathrm{E}_x\left[(g^{\mathcal{D}}(x) - f(x))^2\right]\right] \\ &= \mathrm{E}_x\left[\mathrm{E}_{\mathcal{D}}\left[(g^{\mathcal{D}}(x) - f(x))^2\right]\right] \end{aligned}$$

since the expected value is a linear operator, and where:

$$E_{\mathcal{D}}\left[(g^{\mathcal{D}}(x)-f(\mathbf{x}))^2\right]=\int(g^{\mathcal{D}}(x)-f(x))^2p(\mathcal{D})d\mathcal{D}.$$

Notice that taking the expectation concerning all data sets eliminates dependence on any particular data set.

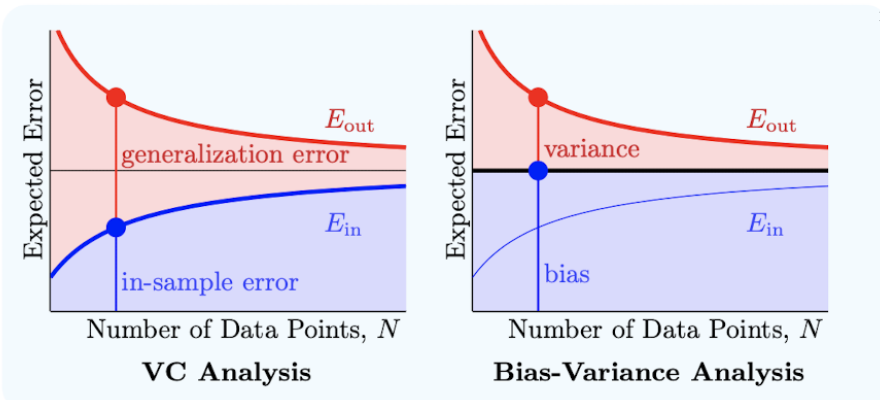


Figure 3: Learning curves. From Learning from Data.

In general for a deterministc function $f(\mathbf{x})$, the bias variance decomposition is:

$$\begin{aligned}
\mathrm{E}_{\mathcal{D}}\left[\mathrm{E}_{\mathsf{out}}(g^{\mathcal{D}})\right] &= \mathrm{E}_{\mathcal{D}}\left[\mathrm{E}_{\mathbf{x}}\left[(g^{\mathcal{D}}(\mathbf{x})-f(\mathbf{x}))^2\right]\right] \\
&= \mathrm{E}_{\mathbf{x}}\left[\mathrm{E}_{\mathcal{D}}\left[(g^{\mathcal{D}}(\mathbf{x})-f(\mathbf{x}))^2\right]\right] \\
&= \mathrm{E}_{\mathbf{x}}\left[\mathrm{E}_{\mathcal{D}}\left[g^{\mathcal{D}}(\mathbf{x})^2\right]-2\underbrace{E_{\mathcal{D}}\left[g^{\mathcal{D}}(\mathbf{x})\right]}_{\text{average function over }\mathcal{D}}f(\mathbf{x})+f(\mathbf{x})^2\right] \\
&= \mathrm{E}_{\mathbf{x}}\left[\mathrm{E}_{\mathcal{D}}\left[g^{\mathcal{D}}(\mathbf{x})^2\right]-2\bar{g}(\mathbf{x})f(\mathbf{x})+f(\mathbf{x})^2\right] \\
&= \mathrm{E}_{\mathbf{x}}\left[\mathrm{E}_{\mathcal{D}}\left[g^{\mathcal{D}}(\mathbf{x})^2\right]-\bar{g}(\mathbf{x})^2+\bar{g}(\mathbf{x})^2-2\bar{g}(\mathbf{x})f(\mathbf{x})+f(\mathbf{x})^2\right] \\
&= \mathrm{E}_{\mathbf{x}}\left[\mathrm{E}_{\mathcal{D}}\left[g^{\mathcal{D}}(\mathbf{x})^2\right]-\bar{g}(\mathbf{x})^2+(\bar{g}(\mathbf{x})-f(\mathbf{x}))^2\right] \\
&= \mathrm{E}_{\mathbf{x}}\left[\mathrm{E}_{\mathcal{D}}\left[(g^{\mathcal{D}}(\mathbf{x})-\bar{g}(\mathbf{x}))^2\right]+(\bar{g}(\mathbf{x})-f(\mathbf{x}))^2\right] \\
&= \mathrm{E}_{\mathbf{x}}\left[\underbrace{\mathrm{E}_{\mathcal{D}}\left[(g^{\mathcal{D}}(\mathbf{x})-\bar{g}(\mathbf{x}))^2\right]}_{\mathsf{variance}(\mathrm{x})}\right]+\mathrm{E}_{\mathbf{x}}\left[\underbrace{(\bar{g}(\mathbf{x})-f(\mathbf{x}))^2}_{\mathsf{bias}(\mathrm{x})}\right] \\
&= \mathrm{E}_{\mathbf{x}}\left[\mathsf{variance}(\mathbf{x})\right]+\mathrm{E}_{\mathbf{x}}\left[\mathsf{bias}(\mathbf{x})\right] \\
&= \mathsf{bias}+\mathsf{variance}.
\end{aligned}$$

When M data sets are considered for learning, each data set will

lead to a particular solution $g_m(x)$ and, therefore:

$$\bar{g}(\mathbf{x}) \approx \frac{1}{M} \sum_{m=1}^{M} g_m(\mathbf{x}).$$

This decomposition highlights the trade-off between bias and variance, which is crucial in model selection and regularization.

Box 1: Example $t(x) = f(x) + \epsilon(x)$

The bias-variance decomposition provides a framework to analyze the error of a learning algorithm by separating it into three components: bias, variance, and irreducible error.
Let us derive it for the function $t(x) = f(x) + \epsilon(x)$, where $f(x)$ is the deterministic (true) function; and $\epsilon = \epsilon(x)$ is Gaussian noise with zero mean and variance σ^2. Now suppose we use a model $g(x)$ to estimate $f(x)$. The prediction error for a given x is measured as the mean squared error (MSE) of $g(x)$ with respect to the true target $t(x)$:

$$\begin{aligned}\text{MSE} &= \mathrm{E}_{\mathrm{x}}\left[(t - g(x))^2\right] \\ &= \mathrm{E}_{\mathrm{x}}\left[(f(x) + \epsilon - g(x))^2\right].\end{aligned}$$

Taking the expectation wrt $p(\mathcal{D})$ and expanding the square:

$$\mathrm{E}_{\mathcal{D}}\left[\text{MSE}\right] = \mathrm{E}_{\mathcal{D}}\left[\mathrm{E}_{\mathrm{x}}\left[(f(x) - g(x))^2 + 2\epsilon(f(x) - g(x)) + \epsilon^2\right]\right].$$

Since ϵ is independent of $g(x)$, and the mean is zero, the cross-term vanishes: $\mathrm{E}_{\mathrm{x}}[\epsilon(f(x) - g(x))] = 0$.
This simplifies to:

$$\mathrm{E}_{\mathcal{D}}\left[\text{MSE}\right] = \mathrm{E}_{\mathcal{D}}\left[\mathrm{E}_{\mathrm{x}}\left[(f(x) - g(x))^2\right] + \mathrm{E}_{\mathrm{x}}[\epsilon^2]\right].$$

Recalling that $\mathrm{E}_{\mathrm{x}}[\epsilon^2] = \sigma^2$, we have:

$$\begin{aligned}\mathrm{E}_{\mathcal{D}}\left[\text{MSE}\right] &= \mathrm{E}_{\mathcal{D}}\left[\mathrm{E}_{\mathrm{x}}\left[(f(x) - g(x))^2\right] + \sigma^2\right] \\ &= \mathrm{E}_{\mathcal{D}}\left[\mathrm{E}_{\mathrm{x}}\left[(f(x) - g(x))^2\right]\right] + \sigma^2 \\ &= \mathrm{E}_{x}\left[\mathrm{E}_{\mathcal{D}}\left[(f(x) - g(x))^2\right]\right] + \sigma^2.\end{aligned}$$

The term $\mathrm{E}_{\mathcal{D}}\left[(f(x) - g(x))^2\right]$ can be further decomposed. Let $\bar{g}(x) = \mathrm{E}_{\mathcal{D}}[g(x)]$ denote the expected prediction of the model across data sets. Then:

$$\mathrm{E}_{\mathcal{D}}\left[(f(x) - g(x))^2\right] = \left(f(x) - \bar{g}(x)\right)^2 + \mathrm{E}_{\mathcal{D}}\left[(g(x) - \bar{g}(x))^2\right].$$

Here:

- $\left(f(x) - \bar{g}(x)\right)^2$ is the squared bias: it measures how far the average prediction of the model is from the true function $f(x)$, for a given x.
- $\mathrm{E}_{\mathcal{D}}\left[(g(x) - \bar{g}(x))^2\right]$ is the variance: it measures the variability of the model's predictions around its mean prediction, for a given x.

Finally:

$$\begin{aligned}\mathrm{E}_{\mathcal{D}}\left[\text{MSE}\right] &= \mathrm{E}_{\mathrm{x}}\left[\left(f(x) - \bar{g}(x)\right)^2\right] + \mathrm{E}_{\mathrm{x}}\left[\mathrm{E}_{\mathcal{D}}\left[(g(x) - \bar{g}(x))^2\right]\right] + \sigma^2 \\ &= \mathsf{bias} + \mathsf{variance} + \sigma^2.\end{aligned}$$

Expectation of a Random Variable

The **expectation** (or **expected value**) of a random variable is a fundamental concept in probability theory that provides a measure of the center of the distribution of the random variable. It is often denoted by $\mathrm{E}[X]$ for a random variable X.

Expectation for a Discrete Random Variable

For a **discrete random variable** X with possible values $x_1, x_2, \ldots$ and corresponding probabilities $p_1, p_2, \ldots$, the expectation is calculated as:

$$\mathrm{E}[X] = \sum_i x_i \cdot p_i$$

where $p_i = P(X = x_i)$.

Expectation for a Continuous Random Variable

For a **continuous random variable** X with probability density function (PDF) $f(x)$, the expectation is calculated as:

$$\mathrm{E}[X] = \int_{-\infty}^{\infty} x \cdot f(x)\, dx$$

Expectation of Several Random Variables

When dealing with multiple random variables, the expectation can be extended to functions of these variables.

Expectation of a Function of Random Variables

Let X and Y be random variables, and let $g(X, Y)$ be a function of X and Y. The expectation of $g(X, Y)$ is given by:

$$\mathrm{E}[g(X,Y)] = \begin{cases} \sum_x \sum_y g(x,y) \cdot P(X = x, Y = y) & \text{(discrete case)} \\ \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x,y) \cdot f_{X,Y}(x,y)\, dx\, dy & \text{(continuous case)} \end{cases}$$

where $f_{X,Y}(x, y)$ is the joint probability density function of X and Y.

Linearity of Expectation

One of the most important properties of expectation is its linearity. For any random variables X and Y, and constants a and b, the following holds:

$$\mathrm{E}[aX + bY] = a\mathrm{E}[X] + b\mathrm{E}[Y]$$

This property holds regardless of whether X and Y are independent or not.

Expectation of Independent Random Variables

If X and Y are independent random variables, then the expectation of their product is the product of their expectations:

$$\mathrm{E}[XY] = \mathrm{E}[X] \cdot \mathrm{E}[Y]$$

This special case arises from the independence of X and Y.

- Expectation of a Single Random Variable: For a discrete random variable, the expectation is the sum of each possible value multiplied by its probability. For a continuous random variable, the expectation is the integral of the value multiplied by its probability density function.
- Expectation of Several Random Variables: The expectation can be extended to functions of multiple random variables, either through summation (for discrete variables) or integration (for continuous variables). The linearity of expectation allows us to compute the expectation of linear combinations of random variables efficiently. For independent random variables, the expectation of their product is the product of their expectations.

Properties of Mathematical Expectation

Let X and Y be random variables, and let a and b be constants. The following are the key properties of mathematical expectation:

1. Linearity of Expectation

The expectation operator is linear. For any constants a and b,

$$\mathrm{E}[aX + bY] = a\mathrm{E}[X] + b\mathrm{E}[Y].$$

This property holds regardless of whether X and Y are independent.

2. Expectation of a Constant

The expectation of a constant c is the constant itself:

$$\mathrm{E}[c] = c.$$

3. Expectation of a Function of a Random Variable

For any function $g(X)$ of a random variable X,

$$\mathrm{E}[g(X)] = \begin{cases} \sum_x g(x) \cdot P(X = x) & \text{(discrete case)} \\ \int_{-\infty}^{\infty} g(x) \cdot f_X(x)\, dx & \text{(continuous case)} \end{cases}$$

where $f_X(x)$ is the probability density function of X.

4. Expectation of Independent Random Variables

If X and Y are independent random variables, then

$$\mathrm{E}[XY] = \mathrm{E}[X] \cdot \mathrm{E}[Y].$$

This property does not hold if X and Y are dependent.

5. Monotonicity of Expectation

If $X \leq Y$ almost surely (i.e., with probability 1), then

$$\mathrm{E}[X] \leq \mathrm{E}[Y].$$

6. Expectation of a Product of Random Variables

For any two random variables X and Y,

$$\mathrm{E}[XY] = \mathrm{E}[X]\mathrm{E}[Y] + \mathrm{Cov}(X, Y),$$

where $\mathrm{Cov}(X, Y) = \mathrm{E}[(X - \mathrm{E}[X])(Y - \mathrm{E}[Y])]$ is the covariance of X and Y.

7. Expectation of a Sum of Random Variables

For any two random variables X and Y,

$$\mathrm{E}[X + Y] = \mathrm{E}[X] + \mathrm{E}[Y].$$

This property extends to any finite number of random variables.

8. Expectation of a Conditional Expectation

The law of iterated expectation states that

$$\mathrm{E}[X] = \mathrm{E}[\mathrm{E}[X \mid Y]],$$

where $\mathrm{E}[X \mid Y]$ is the conditional expectation of X given Y.

9. Non-Negativity of Expectation

If $X \geq 0$ almost surely, then

$$\mathrm{E}[X] \geq 0.$$

10. Jensen's Inequality

If ϕ is a convex function, then

$$\phi(\mathrm{E}[X]) \leq \mathrm{E}[\phi(X)].$$

If ϕ is concave, the inequality is reversed:

$$\phi(\mathrm{E}[X]) \geq \mathrm{E}[\phi(X)].$$

Excercise

Assume that X and Y are independent. Show that

$$\begin{aligned} \mathrm{E}[g(X,Y)] &= \mathrm{E}_Y[\mathrm{E}_X[g(X,Y)]] \\ &= \mathrm{E}_X[\mathrm{E}_Y[g(X,Y)]]. \end{aligned}$$